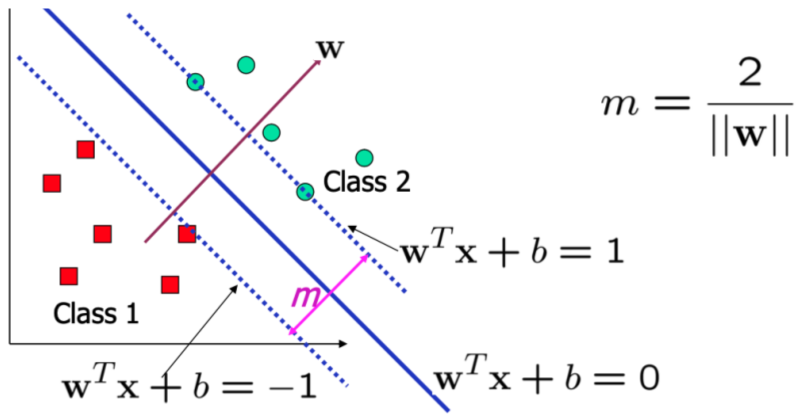
SVM

Support Vector Machine (SVM) is a supervised learning algorithm used for **binary classification.** The **goal** is to find a **decision boundary** (hyperplane) that best separates two classes **with the maximum margin**.



Support vectors are the **data points that are closest to the decision boundary** (hyperplane). These points are **critical** because they **define** the position and orientation of the decision boundary.

The **margin (m)** is the distance between two lines (support vectors). SVM **maximizes this margin (minimize ||w||)** while ensuring all training points are **on the correct side** of the margin.

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad \forall i$$

(Ensures that **each data point** is correctly classified **and lies outside the margin**.)

In this way, we are finding the **widest possible margin** (small ‖w‖) that still keeps **all points correctly classified** and **outside the margin boundaries**.

SVMs can be extended to **nonlinear classification** using the **kernel trick**.